

Uniform Circular Motion

Using polar coordinates, let us track an object $\mathbf{r}(t) = r(t)\hat{\mathbf{r}}(t) \in \mathbb{R}^2$ that satisfies the following:

$$\begin{aligned}\mathbf{r}(t) &= R\hat{\mathbf{r}}(t), \text{ for some constant } R \in \mathbb{R}^+, \\ |\dot{\mathbf{r}}(t)| &= v, \text{ for some constant } v \in \mathbb{R}^+.\end{aligned}$$

The first condition constrains our trajectory to a circle of radius R .

The second condition ensures our object travels around its trajectory with constant speed.

Thus we have set up a **uniform circular motion** problem.

Acceleration

We want to find an expression for $a := |\ddot{\mathbf{r}}|$, the magnitude of the object's acceleration. We differentiate twice.

$$\begin{aligned}\dot{\mathbf{r}} &= \frac{d}{dt}(R\hat{\mathbf{r}}) = R\frac{d\hat{\mathbf{r}}}{dt} = R\frac{d\theta}{dt}\hat{\boldsymbol{\theta}}. \\ \ddot{\mathbf{r}} &= \frac{d}{dt}\left(R\frac{d\theta}{dt}\hat{\boldsymbol{\theta}}\right) = R\frac{d^2\theta}{dt^2}\hat{\boldsymbol{\theta}} - R\left(\frac{d\theta}{dt}\right)^2\hat{\mathbf{r}}.\end{aligned}$$

Applying our second condition, we see that

$$|\dot{\mathbf{r}}(t)| = \left|R\frac{d\theta}{dt}\hat{\boldsymbol{\theta}}\right| = R\left|\frac{d\theta}{dt}\right| = v,$$

which means that $\frac{d\theta}{dt}$ is constant, say $\frac{d\theta}{dt} = \omega$ for some $\omega \in \mathbb{R}$. Then we see $\frac{d^2\theta}{dt^2} = 0$, and therefore

$$\ddot{\mathbf{r}} = -R\omega^2\hat{\mathbf{r}},$$

which, taking the norm and using $\omega = \frac{v}{R}$, gives

$$\boxed{a = \frac{v^2}{R}}.$$

Circular Planetary Orbit

A system of bodies indexed i , each with position $\mathbf{r}_i$ and mass m_i is described by the Newtonian N-body equation of motion:

$$m_i\ddot{\mathbf{r}}_i = G \sum_{j \neq i} m_i m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|^3}$$

If we consider a simple star-planet system, we obtain

$$\ddot{\mathbf{r}}_{\text{star}} = Gm_{\text{planet}} \frac{\mathbf{r}_{\text{planet}} - \mathbf{r}_{\text{star}}}{|\mathbf{r}_{\text{planet}} - \mathbf{r}_{\text{star}}|^3}$$

$$\ddot{\mathbf{r}}_{\text{planet}} = Gm_{\text{star}} \frac{\mathbf{r}_{\text{star}} - \mathbf{r}_{\text{planet}}}{|\mathbf{r}_{\text{star}} - \mathbf{r}_{\text{planet}}|^3}$$

with $m_{\text{star}} \gg m_{\text{planet}}$, we get that

$$a_{\text{earth}} = \frac{Gm_{\text{sun}}}{R^2},$$

and, imposing uniform circular motion, we get

$$\begin{aligned} \frac{Gm_{\text{star}}}{R^2} &= \frac{v^2}{R} \\ \Rightarrow v &= \sqrt{\frac{Gm_{\text{star}}}{R}}. \end{aligned}$$

This means that if we initialize our planet to a position of distance R from the star and a speed $v = \sqrt{\frac{Gm_{\text{star}}}{R}}$ tangential to the circle of radius R centred at the position of the star, then we get circular motion.

Of course, we are assuming that the position of the star is fixed and does not experience a gravitational force by the planet.